# GARRICK JOHN JENSEN

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## EDUCATION

**University of California, Los Angeles** Expected June 2028
Master of Science, Mechanical Engineering

**Harvey Mudd College** May 2022
Bachelor of Science, General Engineering | Graduated with Distinction, Dean's List

## TECHNICAL SKILLS

**Languages & Tools:** Python, MATLAB, C/C++ (coursework), Linux, Git; PyTorch, Scikit-learn, SciPy, OpenCV
**Robotics & Perception:** Computer vision, segmentation, object detection, pose estimation, sensor fusion, state estimation (EKF, particle filters), occupancy mapping, path planning (RRT, coverage)
**Learning & Signals:** Deep learning, transformers, CNNs, reinforcement learning, world models, multi-agent RL, GPU-accelerated simulation, Fourier analysis, FMCW radar, LiDAR, EO/IR

## RESEARCH AND PROFESSIONAL EXPERIENCE

**UCLA Robotics VECTR Lab, Los Angeles, CA** September 2026–Present
*Graduate Student Researcher*

- Developing a multi-agent RL environment for cooperative search and revisit in GPS- and comms-denied conditions, with simulated LiDAR and occupancy mapping under pose uncertainty
- Using model-based (DreamerV3) RL for embodied communication and teaming via degraded physical signals

**Areté Associates, Northridge, CA** August 2022–Present
*Scientist 2*

- Developed computer vision algorithms and deep learning models for an autonomous triaging device for mass casualty events
- Conducted field tests in military exercises and national competitions such as the DARPA Triage Challenge; presented work at the Military Health System Research Symposium
- Beginning to end training and validation of a joint human pose, body part, and skin segmentation model for real time deployment in austere military conditions
- Signal processing in both the real and Fourier space for heart and respiration rate extraction from FMCW radar and IR/EO imagery for a multimodal vitals monitor
- Design of a custom 2D transformer based classification model for efficient detection of 3D targets in undersea LiDAR, outperforming traditional 3D ResNeXt models with a 0.97 AUC
- Trained RL agents deployed on drones for navigating high dimensional spaces under time constraint

**Doosan Bobcat, West Fargo, ND** August 2021–May 2022
*Capstone Team Lead and Robotics Engineer*

- Developed a coverage path planning algorithm using boustrophedon curves optimized as a modified traveling salesman problem
- Implemented an Extended Kalman Filter fusing radar and GPS for localization, plus a particle filter for obstacle detection and avoidance, on an autonomous mower
- Led a five-person team: set schedule, budget, and tasking, and served as primary liaison to the company sponsor

## SELECTED PROJECTS

**RRT Path Planning for Obstacle Avoidance and Rendezvous.** Rapidly-exploring Random Trees planning to a moving target through cluttered space.

**Two-Segment Robotic Arm.** Dynamically modeled and built a stepper-driven planar arm; programmed an STM32 Nucleo-64 and Max1000 FPGA over SPI and UART to trace keypad-entered shapes.

## PATENT AND SELECTED POSTER PRESENTATIONS

**Patent Pending (named inventor).** Schmidt, M., et al. "Path Planning for Automatic Mowers." WO2023205244A1, Oct. 2023.

**Jensen, G.**, Sato, R., Smith, R., Gushurst, W. "Reinforcement Learning Based Path Planning for Injury-Aware Mass Casualty Monitoring." Military Health System Research Symposium (MHSRS), Aug. 2026.

**Jensen, G.**, Peel, J., Roman, K. "Adapting Open-Source Artificial Intelligence to Combat Casualty Care: Joint Human and Skin Segmentation Neural Network." MHSRS, Aug. 2024.

Four additional co-authored poster presentations at MHSRS (2023, 2024) and BMES (2019).